

Human or Machine? Linguistic Patterns in Human and LLM-Generated Dialogue

Motivation

Large language models are increasingly used in dialogue-based applications such as assistants, tutors, customer-service systems and social chatbots. Their responses are often fluent and appear natural, but it is not clear whether LLM-generated conversations reproduce the linguistic and interactional patterns found in conversations between humans. Human and LLM-generated conversations may differ in relatively superficial properties, such as response length or punctuation, but also in more fundamental characteristics such as syntactic structure, use of questions and acknowledgements, emotional expression, and the extent to which conversational partners adapt their language to one another.

In this project, you will *investigate which linguistic properties distinguish human-human from LLM-generated conversations, whether these properties can be used to detect machine-generated dialogue, and how robust the observed differences are.*

Research questions

Your experiments should address the following questions:

- what basic linguistic and conversational properties differ between human-human and LLM-generated dialogues?
- are there systematic lexical, syntactic, stylistic or interactional differences between the two corpora?
- do humans and LLMs differ in how much they adapt their language to their conversational partner?
- can these linguistic properties be used to distinguish human from LLM-generated dialogue?
- does adding interpretable linguistic features improve over a simple text-based baseline?
- do the observed differences and detection performance remain when superficial stylistic properties of the text are modified?
- do patterns learned from one LLM generalise to dialogue generated by another LLM?

!!! Important !!! Your objective is not simply to obtain the highest classification score. The main objective is to determine what properties distinguish human and LLM dialogue and whether these differences reflect robust conversational behaviour rather than dataset-specific artefacts.

Data

You will receive two related dialogue corpora:

- EmpatheticDialogues (<https://github.com/facebookresearch/EmpatheticDialogues>) containing conversations between human participants
- 2GPTEmpathicDialogues (<https://github.com/morganlee123/2GPTEmpathicDialogues>) containing conversations generated by two independent instances of ChatGPT under scenarios corresponding to those in EmpatheticDialogues

Perform a short exploratory analysis of both datasets. Report (at least):

- number of dialogues
- distribution of turns per dialogue
- distribution of words per turn
- sentence-length statistics
- representative examples

Pay particular attention to simple differences that could make human/LLM classification artificially easy. For example, are LLM responses systematically longer than human responses? Are conversations more regular in length? Are particular expressions or formatting conventions present in only one corpus?

How does human and LLM dialogue differ?

Select several linguistic characteristics and compare their distributions between the two corpora. Your analysis should include features from at least three different levels of linguistic analysis.

Possible lexical features include:

- turn length and its variability
- sentence length
- lexical diversity
- repetition
- contractions
- vocabulary overlap between speakers

Possible syntactic features include:

- part-of-speech distributions
- dependency-tree depth
- dependency distance
- clause density/sentence complexity
- frequency of selected syntactic constructions

Possible stylistic or pragmatic features include:

- questions
- hedging and intensifiers
- politeness/formality
- punctuation
- acknowledgements and backchannels
- sentiment

!!! Important !!! You are not expected to calculate all of these features. Select a small set that allows you to formulate and test interesting hypotheses.

For each selected feature:

- clearly define how it is calculated
- calculate it for each dialogue
- compare its distribution between the human and LLM corpora
- quantify the magnitude of the observed difference
- interpret what the difference might tell us about human and LLM conversational behaviour

Conversational alignment

Human conversational partners often adapt aspects of their language to one another. This phenomenon is generally referred to as *linguistic accommodation or alignment*.

Implement at least one measure of conversational alignment. One possible starting point is Language Style Matching (LSM), which measures similarity in the use of function-word categories between conversational partners.

You may alternatively or additionally investigate:

- lexical overlap between consecutive turns
- semantic similarity between consecutive turns
- similarity in utterance length
- syntactic alignment
- emotional alignment

Calculate your selected alignment measure for each conversation and compare its distribution between human-human and LLM-generated dialogues.

Consider questions such as:

- do LLMs align more or less strongly with their conversational partner than humans do?
- is alignment consistent throughout a conversation?

- does alignment increase as a conversation progresses?
- do different types of alignment (e.g., semantic versus stylistic) produce the same conclusion?

Detecting LLM-generated conversations

Build baseline classifiers that predicts whether a conversation was produced by humans or by LLMs. Report accuracy, precision, recall and macro-F1.

Then construct a second classifier using the linguistic and conversational features calculated in the previous sections. Compare:

- text representation only
- linguistic features only
- text representation + linguistic features

Analyse which features contribute most strongly to the classification. Does adding the linguistic features improve performance? More importantly, which differences appear to be responsible for successful classification?

Are the differences robust?

Investigate whether your conclusions depend on superficial characteristics of the original text. Construct at least one modified version of the evaluation data while attempting to preserve its meaning.

Possible transformations include:

- normalising punctuation and capitalisation
- correcting spelling
- normalising contractions
- controlling for utterance length
- paraphrasing utterances using a language model

Recalculate your most interesting linguistic features and re-evaluate the human/LLM classifier. Compare:

- performance on original dialogue vs. performance on modified dialogue
- linguistic differences before vs. after transformation.

Which differences remain? Which disappear? If classifier performance drops substantially following a relatively meaning-preserving transformation, what does this imply about what the classifier was originally detecting?

!!! Important !!! When applying transformations such as paraphrasing, use the same procedure for both human and LLM-generated dialogue. Otherwise, the transformation procedure itself may introduce a new class-specific signal.

Choose a few particularly interesting results and perform a small error analysis. Look at conversations that:

- are incorrectly classified
- change classification after text transformation/paraphrasing
- receive different predictions from the text-based and linguistic-feature classifiers
- look for recurring phenomena (e.g., unusually long or short responses, questions and follow-up questions, politeness or formality, etc.)

Based on your observations, formulate 2–3 hypotheses about why human and LLM conversations can or cannot be distinguished.

Deliverables

- submit a short report (4–5 pages), together with your code. The report should contain your research questions, a concise description of the datasets and selected linguistic features, the main experimental results, and an analysis of what those results mean

!!! Important !!! Unexpected or negative results are entirely acceptable if they are analysed carefully. The objective is not to demonstrate that LLM-generated text is always detectable, but to understand which linguistic characteristics distinguish human and LLM dialogue, under which conditions those differences appear, and whether they generalise beyond a single dataset (additional datasets if the time permits) or model.